

What is the impact of smoking on mortality risk?

Computational Economics





Context and Motivation

- Smoking remains one of the leading preventable causes of premature death in the U.S.
- Understanding how much smoking increases mortality risk is crucial for economic and health policy.
- Logistic regression provides a clean, interpretable way to quantify the effects of smoking.
- Focusing on a single behavioral factor (smoking) avoids confounding and simplifies analysis.

The goal: measure how smoking status (never, former, current) affects mortality risk while controlling for age, income, and sex.



Research Question

Main question:

How does smoking status affect an individual's mortality risk after accounting for demographic and socioeconomic characteristics?

Secondary interest:

- How much higher is risk for current smokers vs. never smokers?
- Do former smokers still carry elevated risk?
- Does age or income significantly moderate risk?



Repo Set Up

Repository Layout

- `config/`
 - `config.R` : Central configuration (paths, packages, model settings).
 - `requirements.txt` : R package list (informational; the main script installs as needed).
- `utils/` (utility modules)
 - `data_nhanes.R` : Download/read NHANES DEMO/SMQ and mortality raw files.
 - `cleaning.R` : Build feature matrix `X` and label `y` with imputation and standardization.
 - `io.R` : Table/figure writers and report helpers.
 - `reporting.R` : Programmatic LaTeX report generator (`output/report.tex`).
 - `helpers.R` : Small helpers (plot saving, panels, sanitization).
- `core/` (modeling & diagnostics)
 - `modeling.R` : Weighted logistic regression (HC1 robust SEs).
 - `modeling_lasso.R` : Regularized logistic regression (glmnet with CV).
 - `diagnostics.R` : Residual, QQ, and prediction plots.
- `scripts/`
 - `fetch_nhanes_raw.R` : Download raw NHANES and mortality assets into `resources/` .
 - `make_clean_matrices_only.R` : Construct `clean-data/X.csv` , `y.csv` , `cleaned_full.csv` , `meta.csv` and write clean-data LaTeX snippets to `output/tables/` .
- `resources/`
 - `nhanes/` and `mortality/` : Raw input files as downloaded (never overwritten by cleaning).
- `clean-data/`
 - Model-ready matrices (`X.csv` , `y.csv`) and a full cleaned frame (`cleaned_full.csv`).
- `output/`
 - `report.tex` , `report.pdf` : Generated LaTeX and PDF.
 - `report_final.tex` , `report_final.pdf` : Curated final report and its PDF.
 - `tables/` , `figures/` , `artifacts/` : All auxiliary outputs for the report.



Data and Smoker Code

This section describes the data sources, how the files are linked, and the basic sample structure. DEMO and SMQ are joined on the NHANES respondent identifier `SEQN` within the 2013–2014 cycle. Mortality is then linked from the public use National Center for Health Statistics follow up file.

Table 1: Raw NHANES and Mortality Assets

file	wave	years	n	k
DEMO.H.csv	H	2013–2014	10 175.00	47.00
SMQ.H.csv	H	2013–2014	7168.00	32.00
NHANES_2013_2014_MORT_2019_PUBLIC.csv	NA	Mortality 2019 linkage	10 175.00	8.00

Table 1 documents the raw input files, their row counts, and key variables. It shows that the DEMO and SMQ files have similar numbers of observations and that the mortality linkage covers nearly all respondents in the cycle. This confirms that merge keys and file coverage are consistent before any cleaning.

- **SMQ020**: indicator for whether the respondent has smoked at least 100 cigarettes in their life. This defines ever versus never smokers.
- **SMQ040**: indicator for whether the respondent currently smokes cigarettes. This splits ever smokers into current and former smokers.
- If `SMQ020 == 2` (*has not smoked 100 cigarettes*): label as Never.
- If `SMQ020 == 1` and `SMQ040 == 3` (*currently do not smoke*): label as Former.
- If `SMQ020 == 1` and `SMQ040` equals 1 or 2 (*every day or some days*): label as Current.
- Otherwise: smoker is set to NA (insufficient or inconsistent information).



Joined and Cleaned Data

Label variable: dead
Features used: age, sex, income, smoker, smoker_code, age_sq, log_income
Initial n: 6111; **Dropped on label only:** 13; **Final n:** 6098
Columns with missing (pre-impute counts): income, dead

Data source: Cleaned matrices (label = dead) from clean-data/
NHANES cycles: H (2013–2014)
Observations (n): 6098
Variables (k): 8
Total missing cells: 0 (0.00%)

Table 4: Head of Feature Matrix X

age	sex	income	smoker	smoker_code	age_sq	log_income
1.17	male	−0.99	Former	2.00	1.37	0.61
0.36	male	−0.40	Current	3.00	0.13	1.02
1.33	male	1.32	Former	2.00	1.78	1.71
1.39	female	1.62	Never	1.00	1.92	1.79
0.47	male	1.49	Former	2.00	0.22	1.76
0.74	female	1.62	Never	1.00	0.54	1.79

Table 5: Head of Label Vector y

dead
FALSE
TRUE
FALSE
TRUE
FALSE
FALSE

Table 8: Imputation Counts by Variable

variable	n_imputed
age	0.00
income	480.00
sex	0.00
smoker	0.00
smoker_code	0.00



Categorical Metrics

Table 12: Categorical Metrics for Features in X

		N	%
sex	female	3187	52.3
	male	2911	47.7
	Unknown	0	0.0
smoker	Never	3523	57.8
	Former	1344	22.0
	Current	1231	20.2
	Unknown	0	0.0

Table 11: Numeric Metrics for Features in X

	Unique	Missing Pct.	Mean	SD	Min	Median	Max
age	63	0	0.0	1.0	−1.6	0.0	1.8
income	425	0	0.0	1.0	−1.5	−0.3	1.6
smoker_code	3	0	1.6	0.8	1.0	1.0	3.0
age_sq	63	0	1.0	0.9	0.0	0.7	3.1
log_income	425	0	1.1	0.5	0.0	1.1	1.8

Table 13: Outcome Distribution for y

y	count
FALSE	5632.00
TRUE	466.00



More

Table 7: Head of Meta Data (IDs and Missingness Flags)

id	wave	age_missing	income_missing	smoker_code_missing
73 557.00	H	0.00	0.00	0.00
73 558.00	H	0.00	0.00	0.00
73 559.00	H	0.00	0.00	0.00
73 561.00	H	0.00	0.00	0.00
73 562.00	H	0.00	0.00	0.00
73 564.00	H	0.00	0.00	0.00

Table 9: Feature Dictionary for X

column	description
age	Age (years), standardized
sex	Sex factor: female/male
income	Income-to-poverty ratio (PIR), standardized
smoker	Smoking status factor: Never/Former/Current from SMQ020/040
smoker_code	Ordinal smoking code: Never=1, Former=2, Current=3; 0 if imputed
age_sq	Age squared, from standardized age
log_income	log1p(PIR) from imputed income

Variance Inflation Factors (VIF).

	GVIF	Df	$GVIF^{(1/(2 \cdot Df))}$
smoker	1.179329	2	1.042098
age	1.109948	1	1.053541
age_sq	1.049561	1	1.024481
sex	1.035840	1	1.017762
log_income	1.075912	1	1.037262



Class Imbalance $\rightarrow$ Weights

Let $\pi = P(y = 1)$ denote the prevalence of deaths. When y is imbalanced (small π), unweighted logistic loss tends to prioritize the majority class. A common remedy is inverse-prevalence weights: $w_1 = 1/\pi$ for positives and $w_0 = 1/(1 - \pi)$ for negatives, often normalized so the average weight equals one. The weighted log-likelihood is

$$L(\beta) = - \sum_i w_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)], \quad p_i = \text{logit}^{-1}(X_i \beta).$$

quantity	value
n_total	6098.00
n_positive (y=1)	466.00
n_negative (y=0)	5632.00
prevalence (y=1)	0.08
w1 = 1/prevalence	13.09
w0 = 1/(1-prevalence)	1.08
w1_normalized	6.54
w0_normalized	0.54



Model 1 (Logit (Weighted, HC1 Robust)):

$$\Pr(y_i = 1 \mid X_i) = \text{logit}^{-1}(\beta_0 + \beta_1 \cdot 1\{\text{smoker}_i = \text{Former}\} + \beta_2 \cdot 1\{\text{smoker}_i = \text{Never}\} + \beta_3 \text{age}_i + \beta_4 1\{\text{sex}_i = \text{male}\} + \beta_5 \log_income_i)$$

Coefficients (robust SEs):

	title
(Intercept)	0.267 (0.193)
smokerFormer	-0.540*** (0.184)
smokerNever	-1.142*** (0.188)
age	1.303*** (0.095)
age_sq	0.451*** (0.069)
sexmale	0.174 (0.128)
log_income	-0.921*** (0.138)
Num.Obs.	6098
RMSE	0.39
Std.Errors	Custom

* p <0.1, ** p <0.05, *** p <0.01

Classification metrics (in-sample, training set)

Metric	Value
Accuracy	0.78
LogLoss	0.46
Brier	0.15
ROC_AUC	0.87
PR_AUC	0.38

Classification metrics (out-of-sample, validation set)

Metric	Value
Accuracy	0.79
LogLoss	0.46
Brier	0.15
ROC_AUC	0.88
PR_AUC	0.39



Model 2 (Regularized Logit (CV, Weighted)):

$$\hat{\beta} = \arg \min_{\beta} \left[-\frac{1}{n} \sum_i w_i (y_i \log p_i + (1 - y_i) \log(1 - p_i)) + \lambda \|\beta\|_1 \right]$$

Nonzero coefficients at the selected penalties:

lambda	term	estimate
lambda.min	(Intercept)	0.24
lambda.min	smokerFormer	-0.50
lambda.min	smokerNever	-1.11
lambda.min	age	1.29
lambda.min	age_sq	0.44
lambda.min	sexmale	0.16
lambda.min	log_income	-0.91
lambda.1se	(Intercept)	-0.13
lambda.1se	smokerNever	-0.54
lambda.1se	age	1.09
lambda.1se	age_sq	0.32
lambda.1se	log_income	-0.60

Classification metrics (in-sample, training set)

Metric	Value
Accuracy	0.78
LogLoss	0.46
Brier	0.15
ROC_AUC	0.87
PR_AUC	0.38

Classification metrics (out-of-sample, validation set)

Metric	Value
Accuracy	0.79
LogLoss	0.46
Brier	0.15
ROC_AUC	0.88
PR_AUC	0.39

Model 2 Cont.

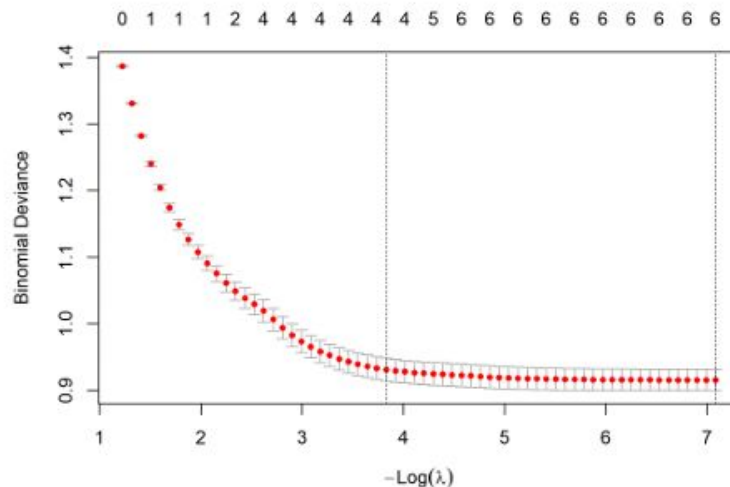


Figure 9: Regularized Logit: Cross Validation Error vs Lambda

Interpretation. The cross-validation error curve is relatively flat near its minimum, indicating that a range of λ values deliver very similar validation performance. The $\lambda_{\min}$ choice gives the smallest error, while the one-standard-error rule selects a slightly larger λ that yields a sparser model with almost indistinguishable predictive performance. This stability suggests that the results are not finely tuned to a single penalty value and that the identification of age, smoking, and income as key predictors is robust.

Model 1 vs. Model 2

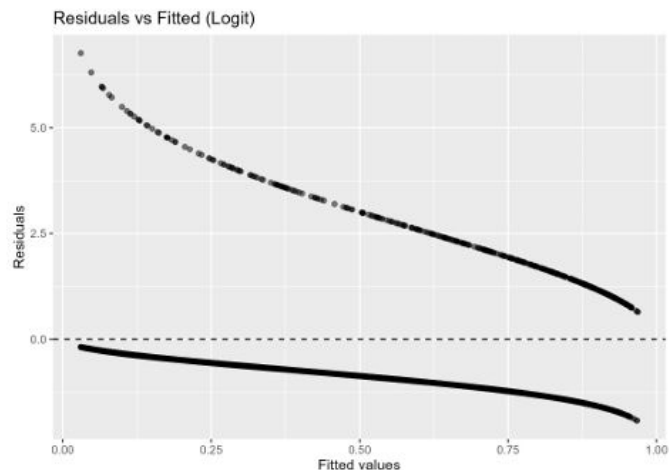


Figure 1: Logit: Residuals vs Fitted (training sample, deviance residuals)

Interpretation. The residuals-versus-fitted plot for the logit shows the familiar “bow tie” pattern associated with binary outcomes: residual variance is highest near fitted probabilities around 0.5 and shrinks toward zero as fitted risks approach 0 or 1. There is no strong evidence of systematic curvature or unmodeled structure in the mean: the residual cloud is roughly centered around zero across the fitted range. This supports the use of a logistic link with the chosen predictors, while reminding us that heteroskedasticity is inherent and motivates the use of robust standard errors.

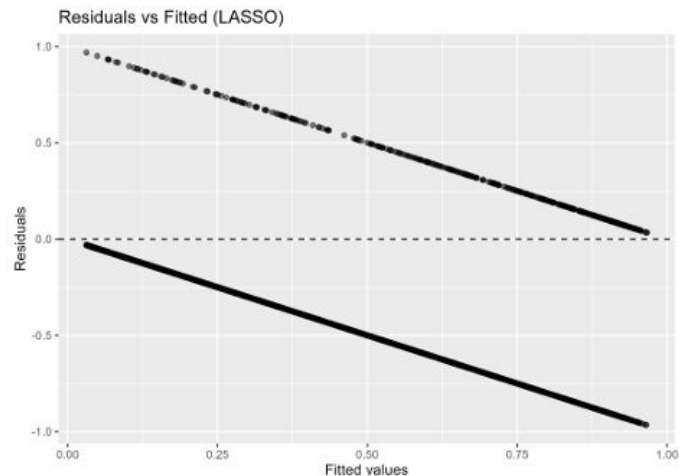


Figure 7: Regularized Logit: Residuals vs Fitted (training sample, $\lambda_{\min}$)

Interpretation. The residuals-versus-fitted plot for the regularized logit at $\lambda_{\min}$ looks very similar to the baseline logit: residual variance peaks around intermediate fitted probabilities and contracts near zero and one. There is no obvious additional structure, confirming that the L_1 penalty does not fundamentally change the mean functional form; it mainly shrinks coefficients without changing the overall shape of the fit.



Cont.

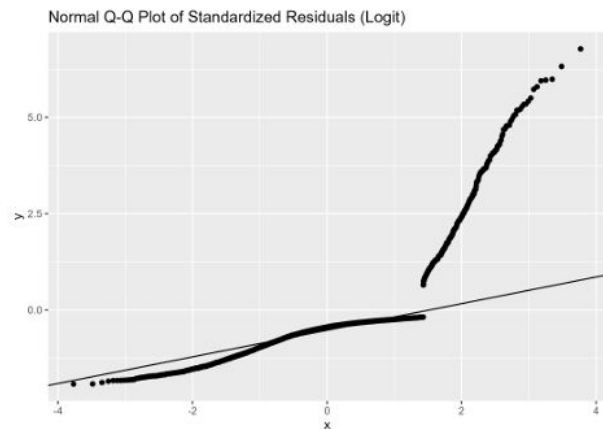


Figure 2: Logit: Normal Q-Q Plot of Deviance Residuals (training sample)

Interpretation. The Q-Q plot for deviance residuals departs from the 45-degree line, especially in the tails, reflecting heavy-tailed behavior relative to a Gaussian reference. This is expected: even after the logit transform, residuals from a binary outcome are not normally distributed. Inference therefore relies on asymptotic theory and robust variance estimates, rather than on exact normality of residuals.

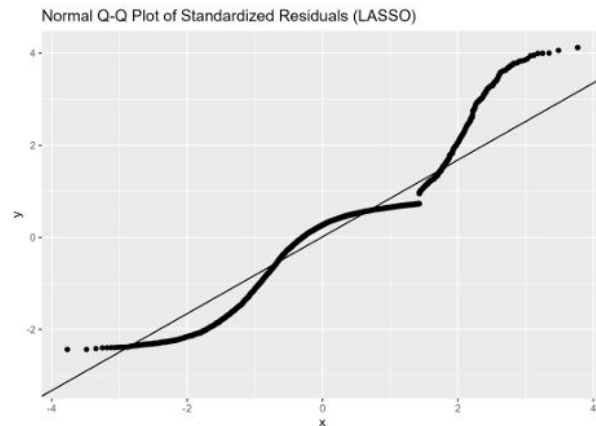


Figure 8: Regularized Logit: Normal Q-Q Plot of Residuals (training sample, $\lambda_{\min}$)

Interpretation. The Q-Q plot for the regularized logit again shows deviations from normality that mirror those of the baseline model, with heavier-than-normal tails. This reinforces the point that Lasso regularization addresses coefficient shrinkage and variable selection, not residual distributional assumptions; robust standard errors and large-sample arguments remain the basis for inference.



Cont.

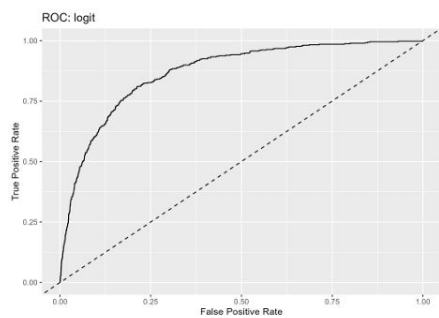


Figure 4: Logit: ROC Curve (out-of-sample, validation set)

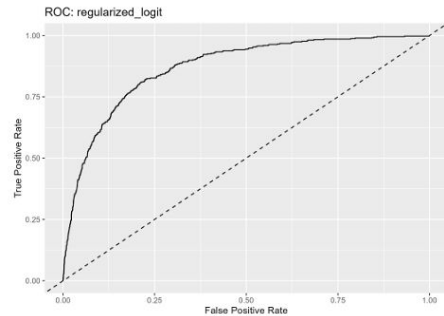


Figure 10: Regularized Logit: ROC Curve (out-of-sample, validation set)

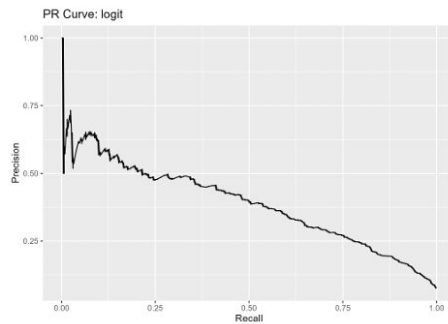


Figure 5: Logit: Precision-Recall Curve (out-of-sample, validation set)

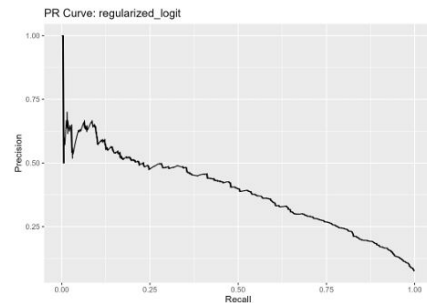


Figure 11: Regularized Logit: Precision-Recall Curve (out-of-sample, validation set)

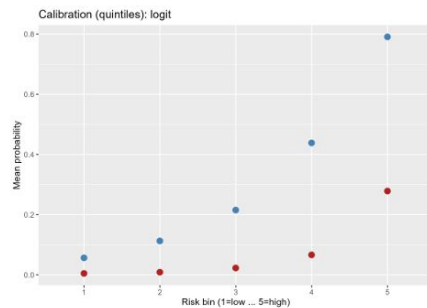


Figure 6: Logit: Calibration by Risk Quintile (out-of-sample, validation set)

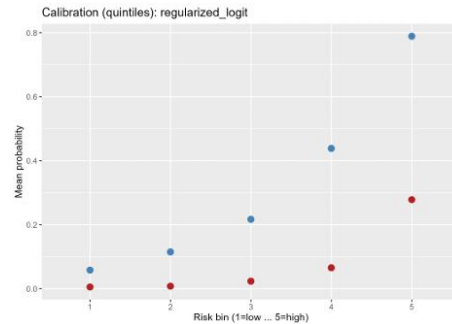


Figure 12: Regularized Logit: Calibration by Risk Quintile (out-of-sample, validation set)



Finally...

Logit vs Regularized Logit (lambda.min and lambda.1se)

term	base_estimate	lasso_min	lasso_1se
(Intercept)	0.27	0.24	-0.13
age	1.30	1.29	1.09
age_sq	0.45	0.44	0.32
log_income	-0.92	-0.91	-0.60
sexmale	0.17	0.16	
smokerFormer	-0.54	-0.50	
smokerNever	-1.14	-1.11	-0.54

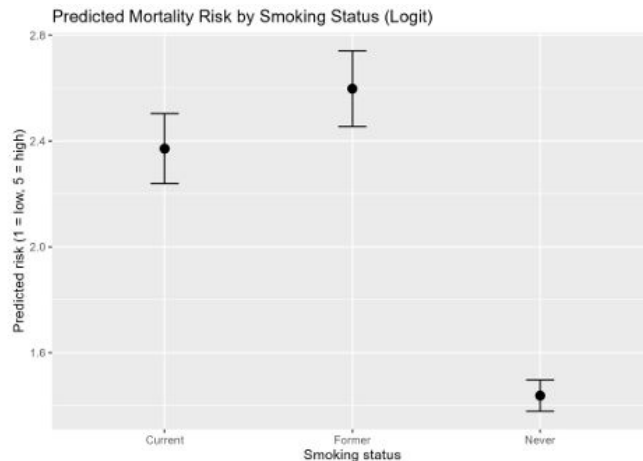


Figure 3: Logit: Predicted Risk by Smoking Status (full analytic sample, scaled 1-5)

Current smokers: 43.71 years (n=1232)
Former smokers: 56.72 years (n=1347)



Results

- Current smokers show a statistically significant increase in mortality risk.
- Former smokers still have higher risk than never smokers AND current smokers (probably because of age, overlapping intervals).
- Age strongly increases risk, consistent and significant.
- Income slightly reduces mortality risk (protective effect).
- Sex may or may not be significant depending on random draw (but included as control).



Policy Implications

Strong evidence that smoking greatly increases mortality risk.

Economic rationale for:

- Higher tobacco taxes
- Targeted cessation programs
- Risk-adjusted insurance pricing
- Public health messaging aimed at younger individuals

QUIT SMOKING EARLY, your age will play apart in longevity



The End